

Vision Transformer

* When trained on mid-size dataset

resnet performance $>$ ViT performance

* When trained on very large
14M - 200M images

resnet $<$ ViT

* So ViT cannot have
come out of an academic lab!

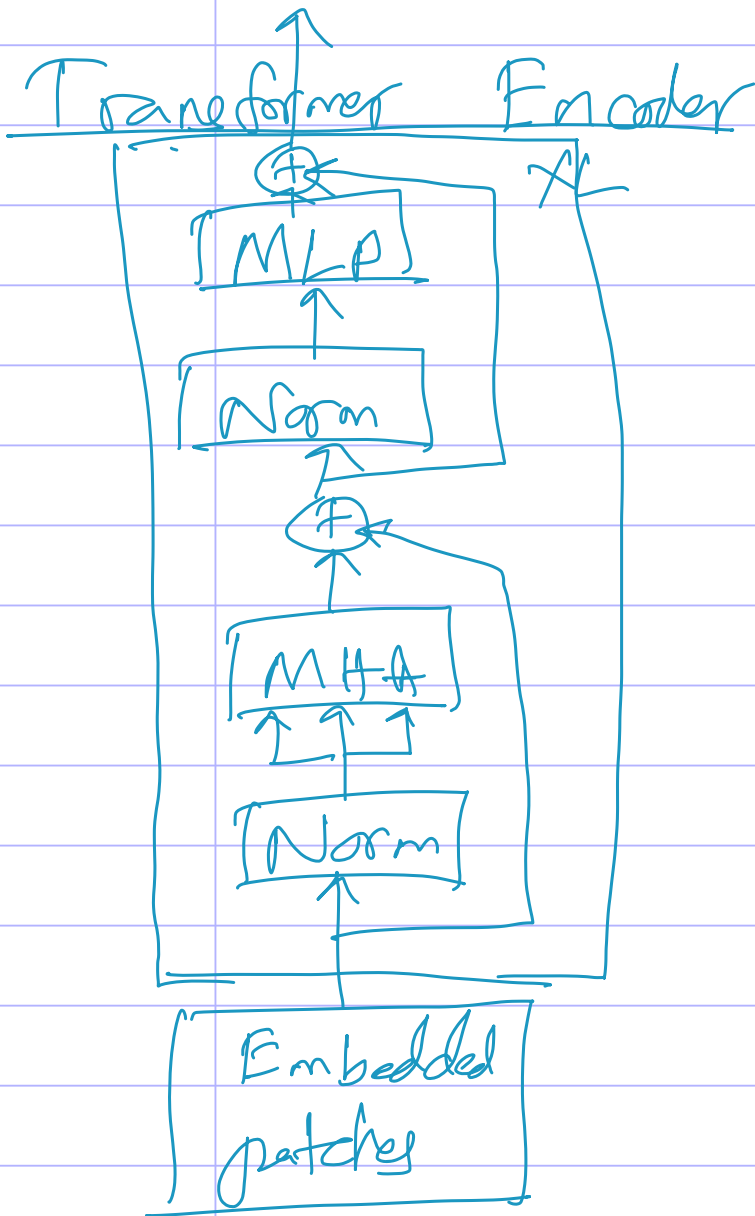
* The training used "IN-HOUSE"
JFT-300M dataset.

$\approx$ So data is also a bottleneck

* I feel like transformers are
easily scalable and trainable

universal funcⁿ approximators

So whoever has high compute & data seem to produce good models.



Multi Head Attention (MHA) Refresher

① Linear projections

$$Q_i = Q W_i^Q$$

$$K_i = K W_i^K$$

$$V_i = V W_i^V$$

~~Q, K, V~~ are matrices from data

~~W~~ W_i^Q, W_i^K, W_i^V are learned projections

② Scaled dot-product attention

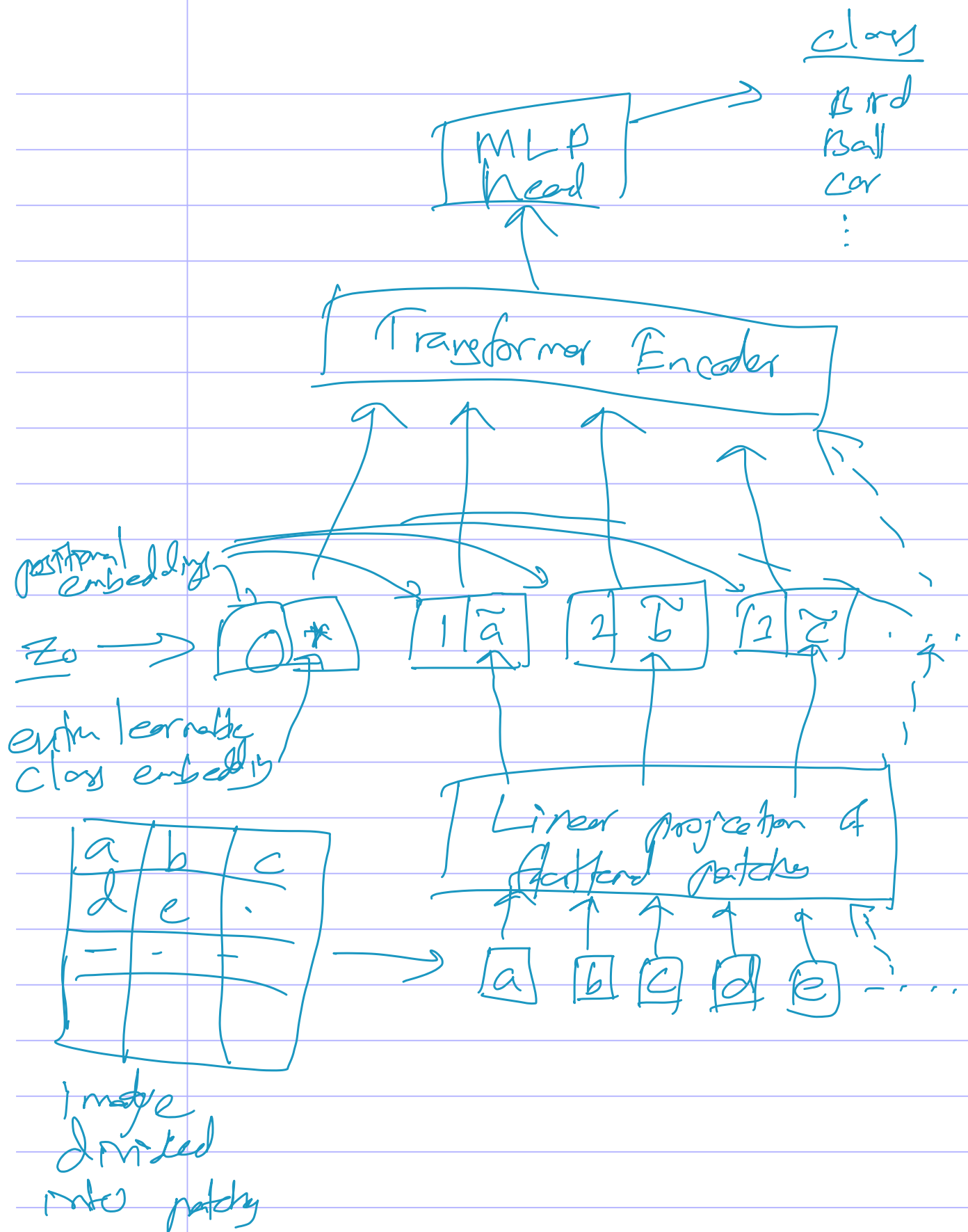
$$\text{head}_i = \text{Attention}(Q_i, K_i, V_i)$$

$$= \text{softmax} \left(\frac{Q_i K_i^T}{\sqrt{d_k}} \right) V_i$$

③ Concatenation & final projection

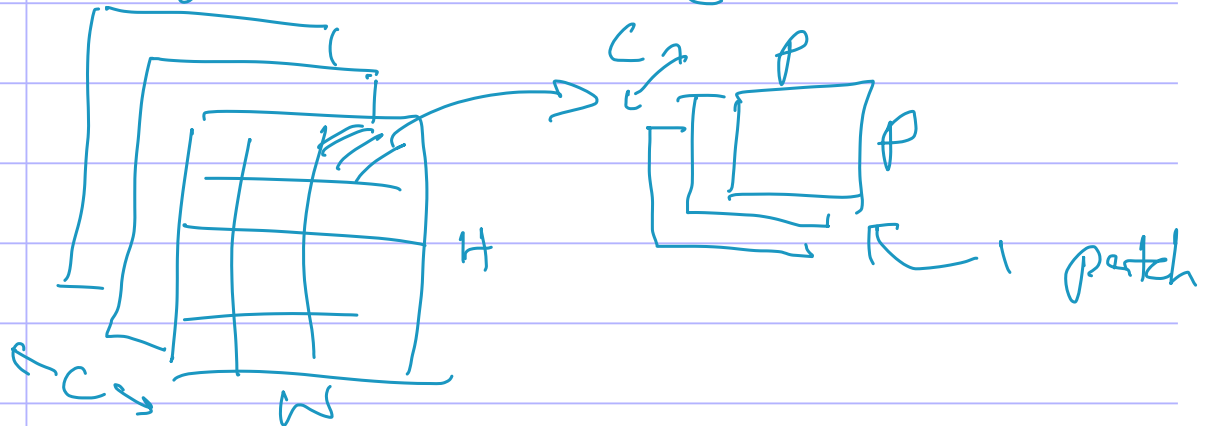
$$\text{Multihead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

ViT architecture



$$\underline{Z}_0 = [\underline{x}_{cls} ; \underline{x}_p^1 E ; \underline{x}_p^2 E ; \dots ; \underline{x}_p^N E] + E_{ps}$$

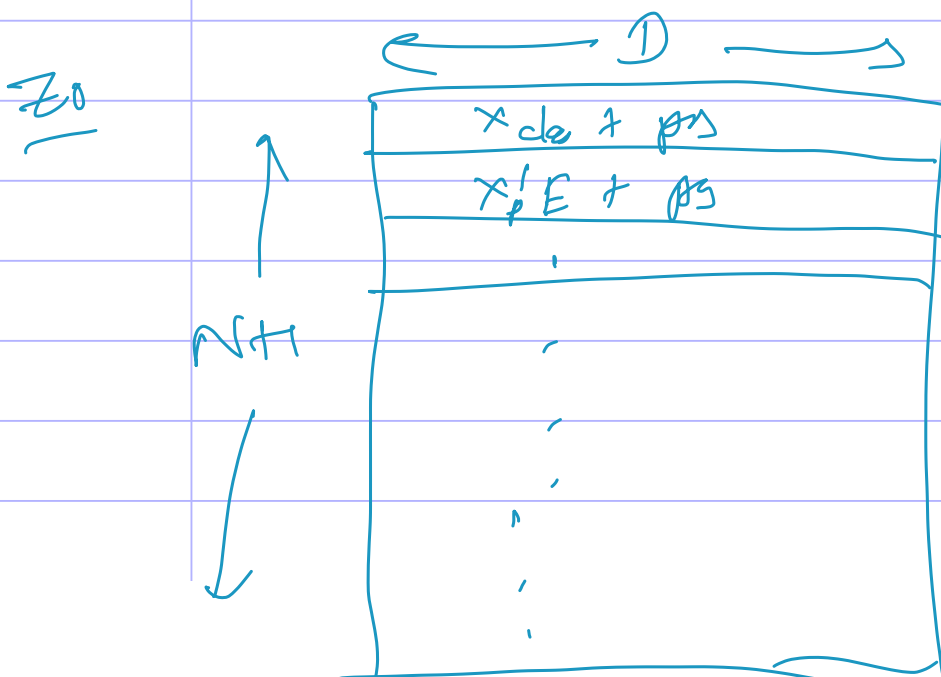
consider image of $H \times W \times C$



$$\# \text{ of patches } N = HW/p^2$$

Embedding matrix $E \in \mathbb{R}^{(p \times C) \times D}$

positional Embedding $E_{ps} \in \mathbb{R}^{(N+1) \times D}$



$$l = 1, \dots, L$$

$$\underline{z}_l' = \text{MLP}(\text{LN}(\underline{z}_{l-1})) + \underline{z}_{l-1}$$

$$\underline{z}_l = \text{MLP}(\text{LN}(\underline{z}_l')) + \underline{z}_l'$$

$$y = \text{LN}(\underline{z}_L^0)$$

* NBK: positional embeddings are just 1D & don't carry any 2D info about the patch location in image.

* Surprise! why destroy information you already have in the image!